

Unit 13

AN INTRODUCTION TO AS LEVEL ORGANIC CHEMISTRY

69 items · 145 marks

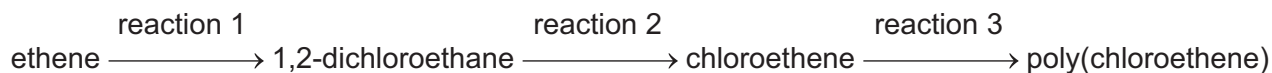
Syllabus sections in this unit

- 13.1 Formulas, functional groups and the naming of organic compounds
- 13.2 Characteristic organic reactions
- 13.3 Shapes of organic molecules; σ and π bonds
- 13.4 Isomerism: structural isomerism and stereoisomerism

Unit 13: An introduction to AS Level organic chemistry**1. M/J 26/P21/Q5(a)**

Context

- 5** Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

**Figure 5.1**

- (a) Complete Table 5.1.

Table 5.1

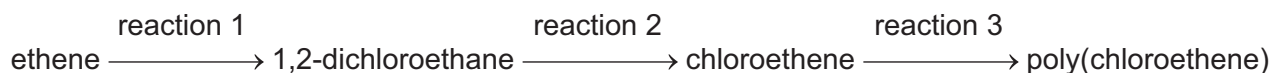
compound	type of hybridisation of carbon atoms
1,2-dichloroethane	
chloroethene	

[2]

2. M/J 26/P21/Q5(c)

Context

- 5** Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

**Figure 5.1**

- (c) Describe what is meant by heterolytic fission of a covalent bond.

[2]

3. M/J 26/P22/Q5(a)

Context

- 5** Hexane, $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_5\text{Cl}$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_3$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_2\text{Cl}$	1,2-dichlorohexane
$\text{CH}_3(\text{CH}_2)_{10}\text{CH}_3$	dodecane
$\text{CH}_3(\text{CH}_2)_3\text{CH}(\text{CH}_3)\text{CH}(\text{CH}_3)(\text{CH}_2)_3\text{CH}_3$	5,6-dimethyldecane

(a) Complete Table 5.1.

[2]

4. M/J 26/P22/Q5(b)(i)-(ii)

Context

- 5** Hexane, $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

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$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_2\text{Cl}$	1,2-dichlorohexane
$\text{CH}_3(\text{CH}_2)_{10}\text{CH}_3$	dodecane
$\text{CH}_3(\text{CH}_2)_3\text{CH}(\text{CH}_3)\text{CH}(\text{CH}_3)(\text{CH}_2)_3\text{CH}_3$	5,6-dimethyldecane

Context: part (b) (introduction)

- (b)** The mechanism for the reaction of hexane with $\text{Cl}_2(\text{g})$ involves three stages.

stage 1 initiation

stage 2 propagation

stage 3 termination

In stage 1 and 2, free radicals are produced.

- (i)** Describe what is meant by free radical.

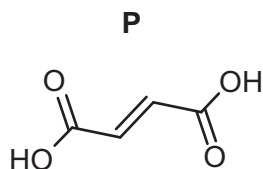
[1]

- (ii)** Name the type of bond breaking that occurs to produce the free radical $\bullet\text{Cl}$ from Cl_2 .

[1]

5. M/J 26/P22/Q6(a)

- 6** **(a)** Figure 6.1 shows the structure of compound **P**.

**Figure 6.1**

- (i)** Draw the structure of the stereoisomer of **P**.

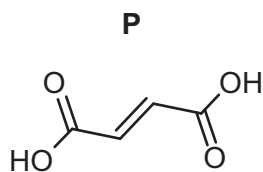
[1]

- (ii)** Explain why **P** shows stereoisomerism.

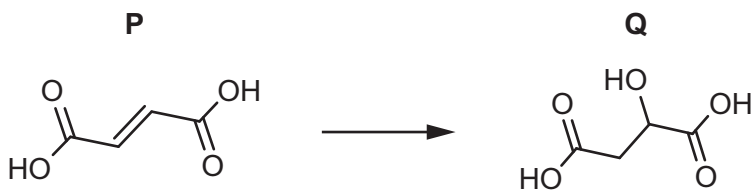
[2]

6. M/J 26/P22/Q6(b)(ii)

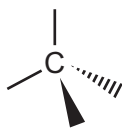
Context: part (a) (introduction)

6 (a) Figure 6.1 shows the structure of compound **P**.**Figure 6.1**

Context: part (b) (introduction)

(b) **P** reacts to produce compound **Q**.**Figure 6.2****(ii)** Complete Figure 6.3 to show the pair of stereoisomers of **Q**.

isomer 1



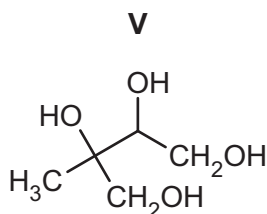
isomer 2

Figure 6.3

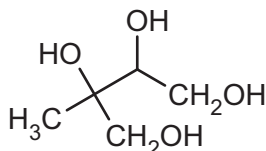
7. M/J 26/P23/Q4(c)(i)-(ii)

Context

- 4** The structure of compound **V** is shown in Figure 4.1.

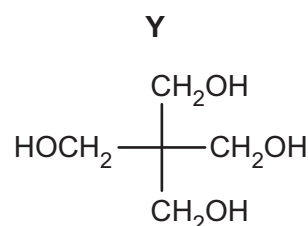
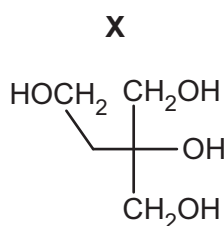
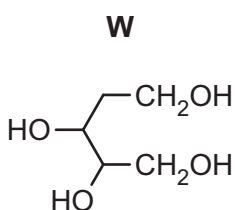
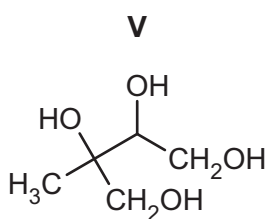
**Figure 4.1**

Context: Fig. 4.1

**Figure 4.1**

Context: part (c) (introduction)

- (c)** Compounds **W**, **X** and **Y** are structural isomers of **V**.

**Figure 4.2**

- (i)** Describe structural isomerism.

[2]

- (ii)** Identify which of structures **V**, **W**, **X** and **Y** show stereoisomerism.

Explain your answer by referring to the relevant parts of the molecule that cause stereoisomerism.

[2]

8. M/J 26/P23/Q5(a)(i)

Context

- 5** Propane, $\text{CH}_3\text{CH}_2\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_2\text{Cl}$	1-chloropropane
$\text{CH}_3\text{CHClCH}_3$	2-chloropropane
$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2,3-dimethylbutane
	hexane
	2-methylpentane

- (a) (i) Complete Table 5.1.

[2]

9. M/J 26/P24/Q5(c)

Context

- 5** Long-chain alkanes from crude oil can be cracked to make alkenes.

- (c) But-1-ene and but-2-ene are structural isomers.

State the type of structural isomerism shown by but-1-ene and but-2-ene.

[1]

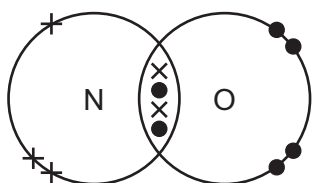
10. MAR 26/P22/Q2(a)(i)

Context

- 2** Nitrogen monoxide, NO, is an atmospheric pollutant.

Context: part (a) (introduction)

- (a) Figure 2.1 shows the dot-and-cross diagram of NO.

**Figure 2.1**

- (i) NO has an unpaired electron.

State the term for a molecule with an unpaired electron.

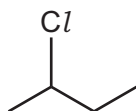
[1]

11. MAR 26/P22/Q3(b)

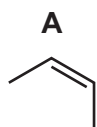
Context

- 3** Secondary halogenoalkanes are important building blocks in organic chemistry.

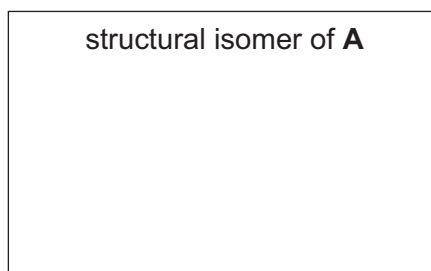
2-chlorobutane is an example of a secondary halogenoalkane.

2-chlorobutane**Figure 3.1**

- (b) 2-chlorobutane can be formed by the electrophilic addition of HCl to different isomers of C_4H_8 . Compound **A** is one of these isomers.

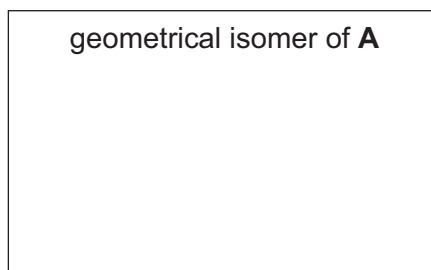
**Figure 3.2**

- (i) Draw a structural isomer of **A** that can also form 2-chlorobutane when it reacts with HCl .



[1]

- (ii) Draw and give the systematic name of the geometrical isomer of **A**.



systematic name

[2]

- (iii) State and explain the origin of geometrical isomerism in alkenes.

[2]

12. MAR 26/P22/Q4(d)

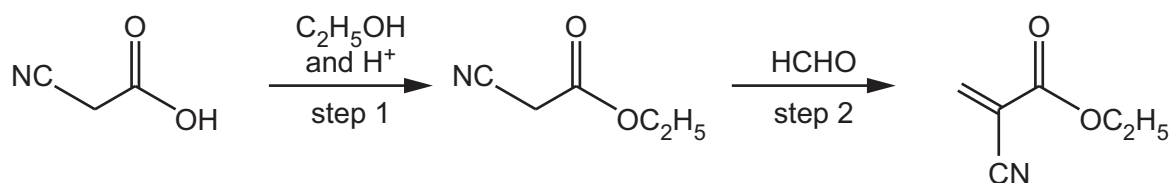
also Unit 22 (2 marks), Unit 18 (1 mark), Unit 20 (1 mark); this unit 2 marks

Context

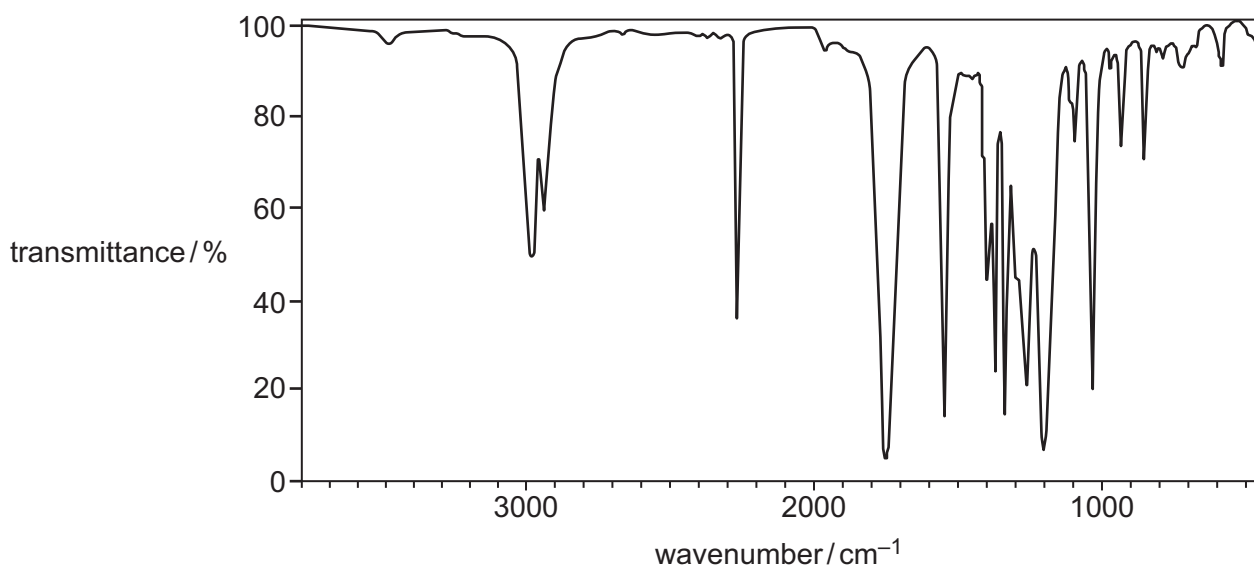
- 4** Cyanoethanoic acid, $\text{CH}_2(\text{CN})\text{COOH}$, is a weak organic acid.

- (d) Figure 4.4 describes a sequence of reactions, starting with $\text{CH}_2(\text{CN})\text{COOH}$, to form **E**.

E

**Figure 4.4**

- (i) Identify the type of reaction that occurs in step 1. [1]
- (ii) Name HCHO , the reactant used in step 2. [1]
- (iii) Deduce the molecular formula of **E**. [1]
- (iv) **E** can be polymerised to form **F**.
Draw the structure of **one** repeat unit of polymer **F**. [1]
- (v) Figure 4.5 shows the infrared spectrum of **E**.

**Figure 4.5****Table 4.1**

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

Using the information given in Table 4.1, identify with a cross (X) on Figure 4.5 an absorption that can be used to monitor the progress of the polymerisation of **E** to form **F**.

Explain your answer with reference to the relevant bond.

[2]

13. O/N 25/P21/Q3(b)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(b) Cyclopentane, C₅H₁₀, has four cyclic structural isomers. One of these isomers is **C**, shown in Fig. 3.1.

Complete Fig. 3.1 to show two other cyclic structural isomers of C₅H₁₀.

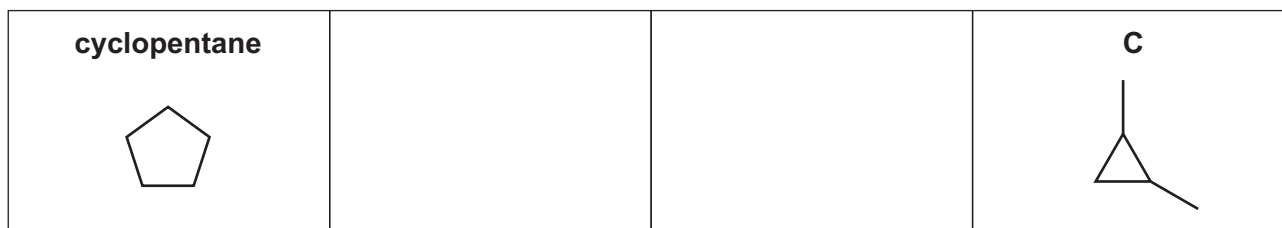


Fig. 3.1

[2]

14. O/N 25/P21/Q3(c)(i)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (c) (introduction)

(c) Cyclopentane reacts with Cl_2 in the presence of ultraviolet light to form $\text{C}_5\text{H}_9\text{Cl}$.

- (i)** The reaction is initiated by the bond fission of Cl_2 .

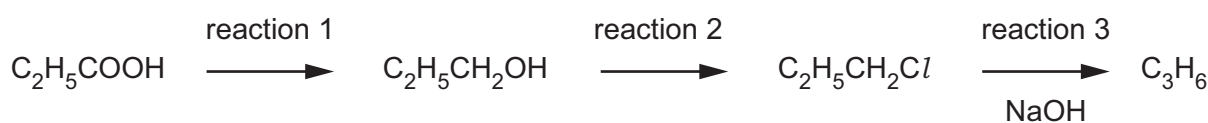
State the type of bond fission shown in the initiation step.

[1]

15. O/N 25/P21/Q4(b)(ii)

Context

- 4** Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

**Fig. 4.1**

Context: part (b) (introduction)

- (b)** Under suitable conditions, propene polymerises to form poly(propene).

Poly(propene) exhibits stereoisomerism.

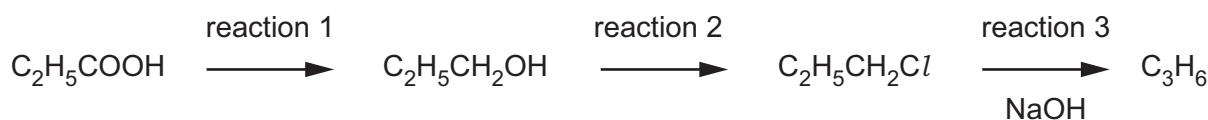
- (ii)** Define stereoisomerism.

[2]

16. O/N 25/P21/Q4(c)(i)

Context

- 4** Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

**Fig. 4.1**

Context: part (c) (introduction)

- (c)** Under different conditions, two molecules of propene can combine to form compounds **X** and **Y**.



- (i)** Name **X**.

[1]

17. O/N 25/P22/Q3(a)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

(a) Define homologous series.

[2]

18. O/N 25/P22/Q3(d)(i)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

Context: part (d) (introduction)

(d) Hexane, C_6H_{14} , has four structural isomers.

Fig. 3.1 shows hexane and two of its structural isomers.

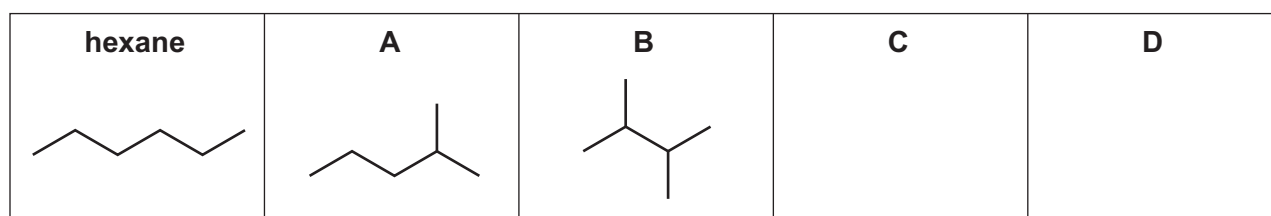


Fig. 3.1

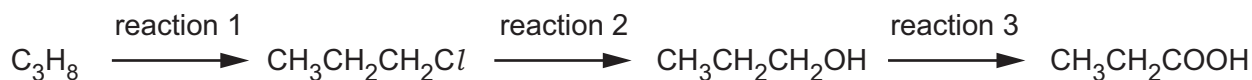
- (i) Complete Fig. 3.1 by drawing structures for **C** and **D**, the other two structural isomers of hexane.

[2]

19. O/N 25/P22/Q4(a)(iii)-(v)

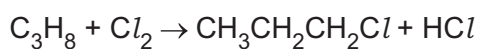
Context

4 Fig. 4.1 shows how propane, C_3H_8 , can be converted to propanoic acid, CH_3CH_2COOH .

**Fig. 4.1**

Context: part (a) (introduction)

(a) Reaction 1 in Fig. 4.1 takes place in the presence of sunlight.



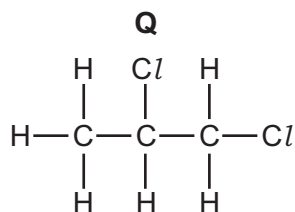
The reaction takes place via initiation, propagation and termination steps.

(iii) Reaction 1 is initiated by the bond fission of Cl_2 .

State the type of bond fission shown in the initiation step.

[1]

(iv) Compound **Q** is a by-product of reaction 1.



Name **Q**.

[1]

(v) The molecular formula of **Q** is $C_3H_6Cl_2$.

Identify the types of structural isomerism and stereoisomerism that a molecule with molecular formula $C_3H_6Cl_2$ can show.

type of structural isomerism

type of stereoisomerism

[2]

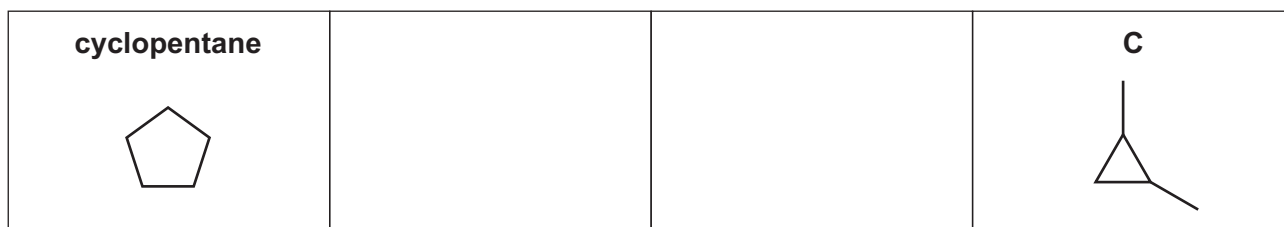
20. O/N 25/P23/Q3(b)

Context

3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(b) Cyclopentane, C_5H_{10} , has four cyclic structural isomers. One of these isomers is **C**, shown in Fig. 3.1.

Complete Fig. 3.1 to show two other cyclic structural isomers of C_5H_{10} .

**Fig. 3.1**

[2]

21. O/N 25/P23/Q3(c)(i)

Context

3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (c) (introduction)

(c) Cyclopentane reacts with Cl_2 in the presence of ultraviolet light to form C_5H_9Cl .

(i) The reaction is initiated by the bond fission of Cl_2 .

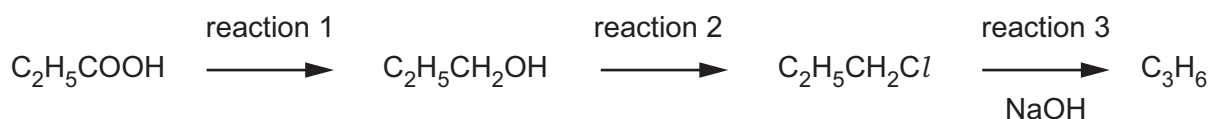
State the type of bond fission shown in the initiation step.

[1]

22. O/N 25/P23/Q4(b)(ii)

Context

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

**Fig. 4.1**

Context: part (b) (introduction)

(b) Under suitable conditions, propene polymerises to form poly(propene).

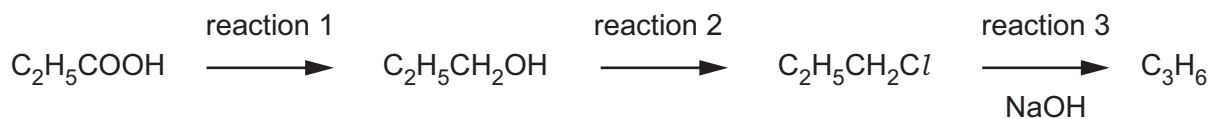
Poly(propene) exhibits stereoisomerism.

(ii) Define stereoisomerism.

[2]

23. O/N 25/P23/Q4(c)(i)

Context

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .**Fig. 4.1**

Context: part (c) (introduction)

(c) Under different conditions, two molecules of propene can combine to form compounds **X** and **Y**.**(i)** Name **X**.

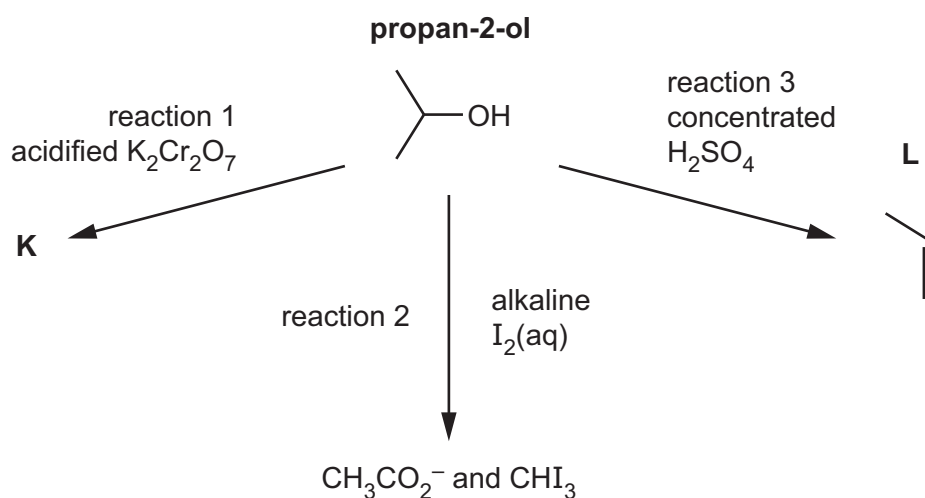
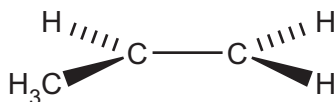
[1]

24. O/N 25/P24/Q3(a)(iv)

Context

3 Propan-2-ol, $(CH_3)_2CHOH$, is sometimes added to fuel to help it burn.

Fig. 3.1 shows some reactions of propan-2-ol.

**Fig. 3.1****(iv)** Complete Fig. 3.2 to show the pi (π) bond in **L** that is formed from orbital overlap.**Fig. 3.2**

[1]

25. M/J 25/P22/Q3(a)(i)

Context: part (a) (introduction)

3 (a) C_3H_6 reacts with HBr(g) in an addition reaction.**(i)** Define addition reaction.

[1]

26. M/J 25/P22/Q4(c)(i)

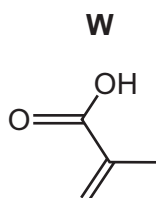
Context: part (c) (introduction)

(c) Compounds **A**, **B** and **C** belong to the alcohol homologous series. Each molecule of **A**, **B** and **C** contains four saturated carbon atoms.**(i)** Identify the type of hybridisation shown in the saturated carbon atoms of all alcohols.

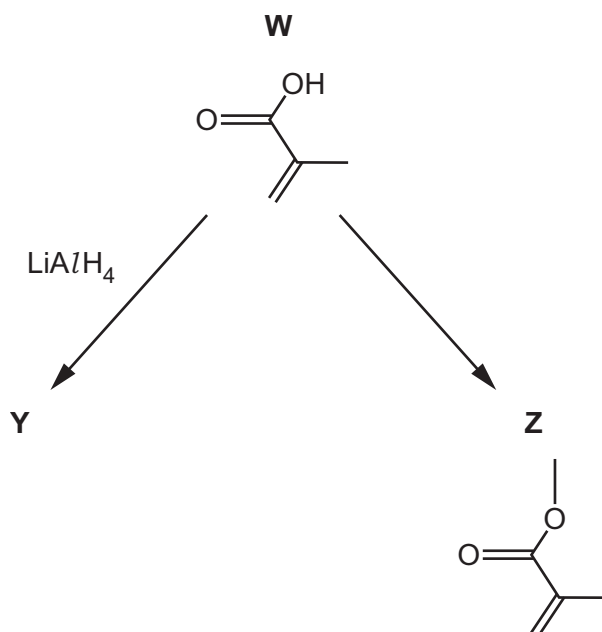
[1]

27. M/J 25/P23/Q5(b)(i)

Context: part (a) (introduction)

5 (a) **W** is a colourless liquid.**Fig. 5.1**

Context: part (b) (introduction)

(b) Fig. 5.2 shows two reactions of **W** to produce organic compounds **Y** and **Z**.**Fig. 5.2****(i)** Deduce the number of sigma (σ) bonds and pi (π) bonds present in **Z**.number of σ bondsnumber of π bonds

[2]

28. M/J 25/P23/Q6(a)(i)

Context: part (a) (introduction)

6 (a) HOCl reacts with $\text{CH}_2=\text{CH}_2$ to produce $\text{HOCH}_2\text{CH}_2\text{Cl}$ in an electrophilic addition reaction.

(i) Define addition reaction.

[1]

29. M/J 25/P24/Q4(a)

Context

4 The skeletal formula of **E** is shown in Fig. 4.1.

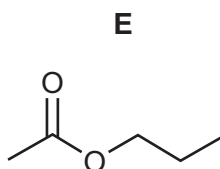


Fig. 4.1

(a) State the molecular formula of **E**.

[1]

30. M/J 25/P24/Q5(a)-(b)

Context

5 The skeletal formula of **W** is shown in Fig. 5.1.

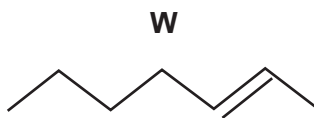
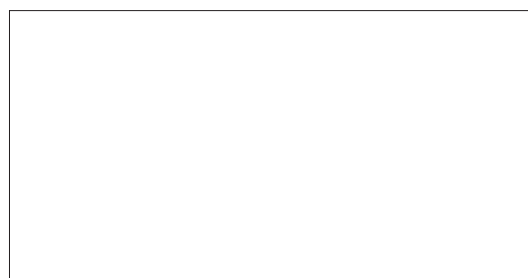
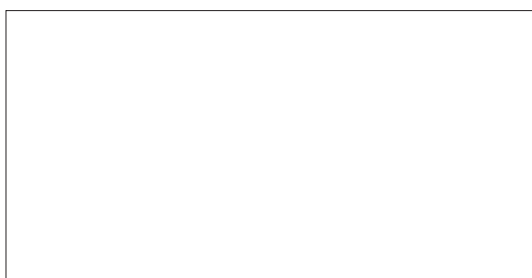


Fig. 5.1

(a) **W** has two positional isomers. Only one shows stereoisomerism.

Draw the structures of the **two** positional isomers of **W** in the boxes.



[2]

(b) (i) Describe the origin of stereoisomerism in **W**.

[1]

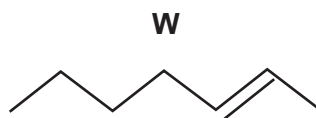
(ii) State why **W** shows stereoisomerism but one of the positional isomers in (a) does **not**.

[1]

31. M/J 25/P24/Q5(c)

Context

- 5** The skeletal formula of **W** is shown in Fig. 5.1.

**Fig. 5.1**

- (c) Draw the skeletal structure of the stereoisomer of **W**.

[1]

32. M/J 25/P24/Q6(a)(iii)

Context

- 6** **P**, **Q** and **R** are three different hydrocarbon molecules.

- (iii) Molecules of **P** have straight chains.

Name the homologous series to which **P** belongs.

[1]

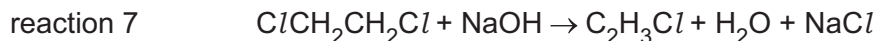
33. MAR 25/P22/Q1(c)(i)

Context

- 1** Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

Context: part (c) (introduction)

- (c) 1,2-dichloroethane, $\text{ClCH}_2\text{CH}_2\text{Cl}$, reacts with NaOH to produce an unsaturated compound, $\text{C}_2\text{H}_3\text{Cl}$, as shown in reaction 7.



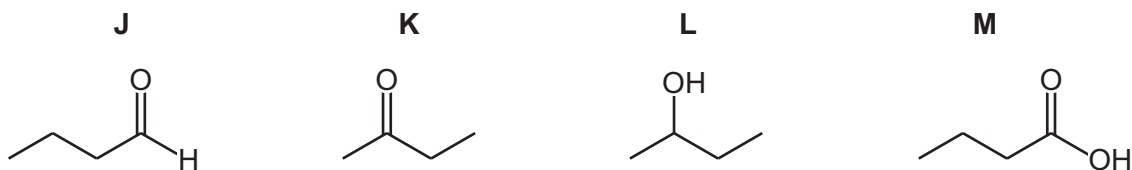
- (i) State what is meant by unsaturated.

[1]

34. MAR 25/P22/Q4(b)(iii)

Context

- 4** Fig. 4.1 shows compounds **J** to **M**, each of which contains four carbon atoms.

**Fig. 4.1**

Context: part (b) (introduction)

- (b) **K** reacts with HCN in the presence of a KCN catalyst, forming **N**.

- (iii) **N** has a chiral centre.

Explain what is meant by a chiral centre.

[1]

35. O/N 24/P21/Q1(d)(ii)-(iv)

Context

- 1** Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

Context: part (d) (introduction)

- (d)** Hydrated rhodium(III) chloride, $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$, catalyses the conversion of ethene to but-2-ene.

Both stereoisomers of but-2-ene are formed in the reaction.

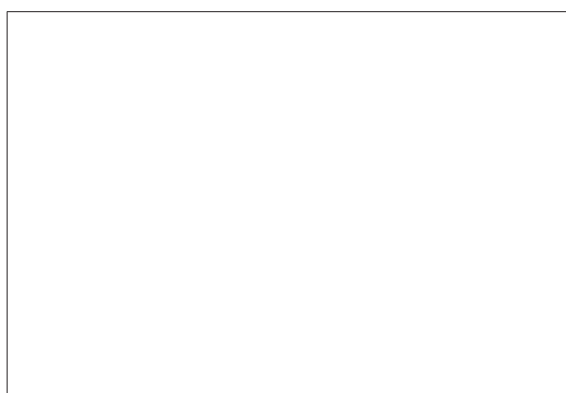
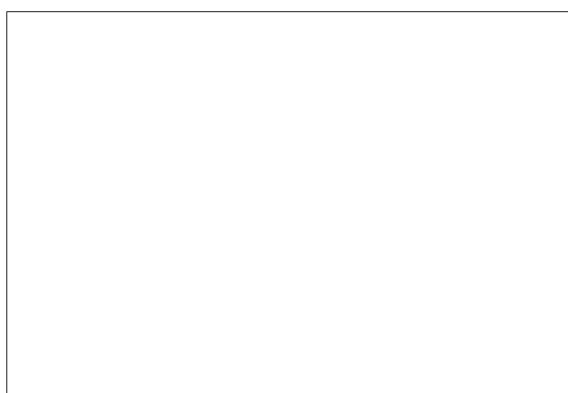
- (ii)** Define stereoisomers.

[1]

- (iii)** Explain how the conversion of ethene to but-2-ene can be described as an addition reaction.

[1]

- (iv)** Draw the two stereoisomers of but-2-ene.



[2]

36. O/N 24/P21/Q2(b)(i)

Context

- 2** Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (b) (introduction)

- (b)** The halogenoalkane $\text{CH}_3\text{CH}_2\text{Cl}$ forms when chlorine reacts with C_2H_6 via a free-radical substitution mechanism.

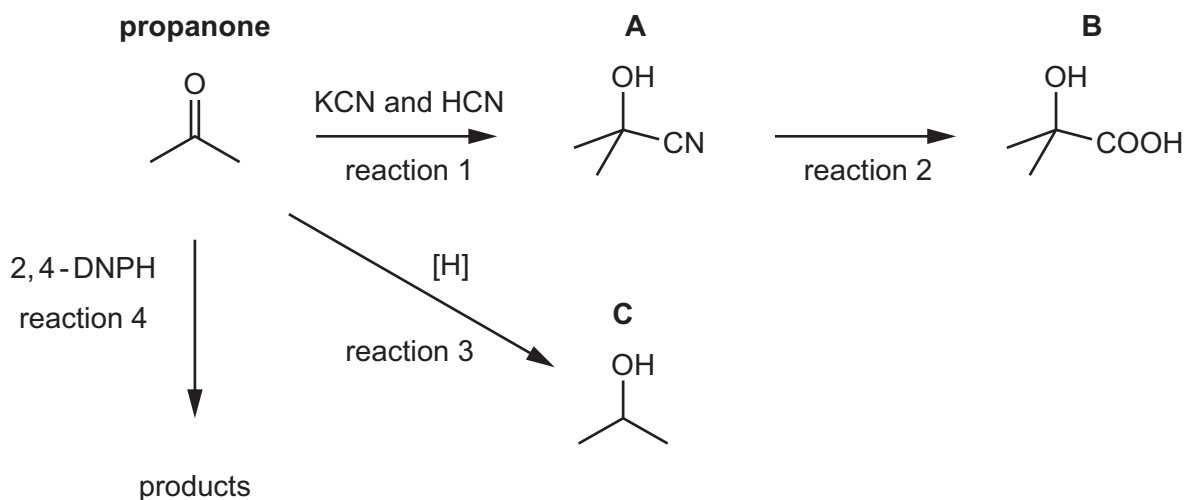
- (i)** Define free radical.

[1]

37. O/N 24/P21/Q4(a)(ii)

Context

- 4** Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

Context: part (a) (introduction)

- (a)** Reaction 1 is a nucleophilic addition reaction.
- (ii)** Explain why **A** does **not** show optical isomerism.

[1]

38. O/N 24/P23/Q1(d)(ii)-(iv)

Context

- 1** Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

Context: part (d) (introduction)

- (d)** Hydrated rhodium(III) chloride, $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$, catalyses the conversion of ethene to but-2-ene.

Both stereoisomers of but-2-ene are formed in the reaction.

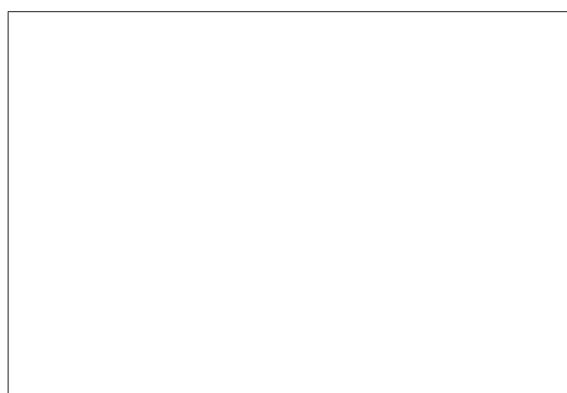
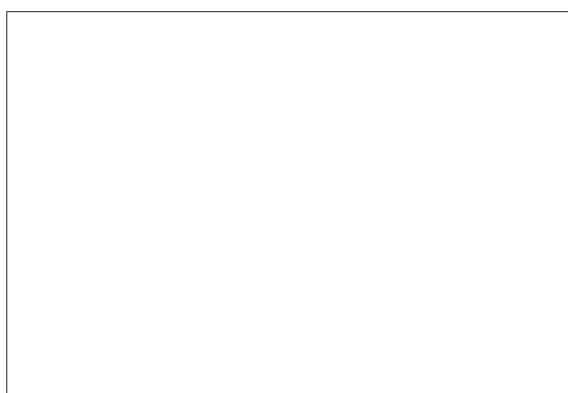
- (ii)** Define stereoisomers.

[1]

- (iii)** Explain how the conversion of ethene to but-2-ene can be described as an addition reaction.

[1]

- (iv)** Draw the two stereoisomers of but-2-ene.



[2]

39. O/N 24/P23/Q2(b)(i)

Context

- 2** Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (b) (introduction)

- (b)** The halogenoalkane $\text{CH}_3\text{CH}_2\text{Cl}$ forms when chlorine reacts with C_2H_6 via a free-radical substitution mechanism.

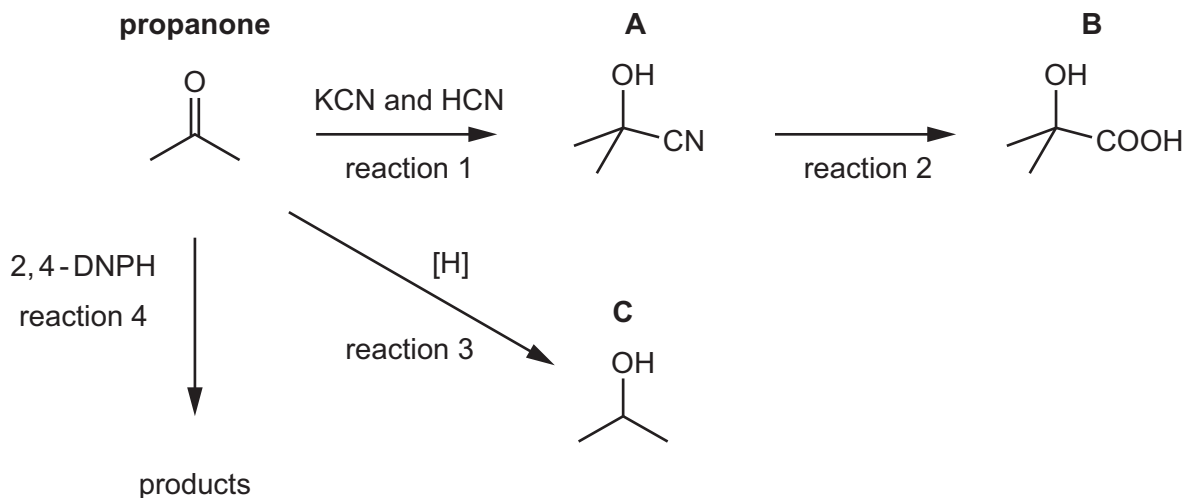
- (i)** Define free radical.

[1]

40. O/N 24/P23/Q4(a)(ii)

Context

- 4** Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

Context: part (a) (introduction)

- (a)** Reaction 1 is a nucleophilic addition reaction.
- (ii)** Explain why **A** does **not** show optical isomerism.

[1]

41. M/J 24/P21/Q5(b)

Context

- 5** Hydrocarbon molecules contain covalent bonds.

- (b)** A $\text{C}=\text{C}$ bond in an alkene is made from a σ bond and a π bond.

- (i)** Identify the hybridisation of the carbon atoms in a $\text{C}=\text{C}$ bond in an alkene.

[1]

- (ii)** Draw labelled diagrams to show, in terms of orbital overlap, how the σ and π bonds are made in a $\text{C}=\text{C}$ bond.

 σ bond π bond

[2]

42. M/J 24/P21/Q5(c)(i)

Context

- 5** Hydrocarbon molecules contain covalent bonds.

Context: part (c) (introduction)

- (c)** In electrophilic reactions involving alkenes the π bond of $\text{C}=\text{C}$ is broken.

- (i)** Suggest **one** difference between σ and π bonds that explains why the π bond is broken in electrophilic addition reactions involving alkenes.

[1]

43. M/J 24/P21/Q6(a)

6 (a) **V** shows stereoisomerism.

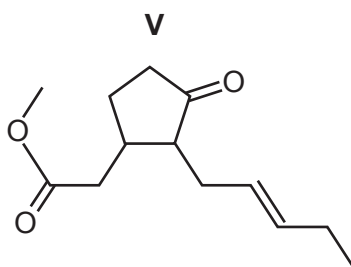


Fig. 6.1

(i) Explain what is meant by stereoisomerism.

[1]

(ii) Deduce the number of stereoisomers of **V**. Explain your reasoning.

[2]

(iii) Deduce the molecular formula of **V**.

[1]

(iv) Name **all** the functional groups present in **V**.

[1]

44. M/J 24/P22/Q4(a)

Context

4 $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ exists as a pair of stereoisomers.

(a) Draw the three-dimensional structures of the **two** stereoisomers of $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$.
R can be used to represent $\text{CH}_3(\text{CH}_2)_5$.

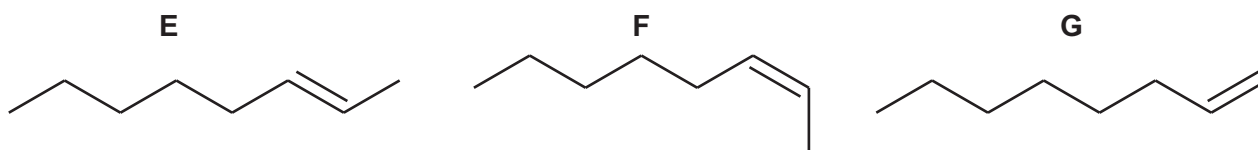
[2]

45. M/J 24/P22/Q4(e)

Context

4 $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ exists as a pair of stereoisomers.

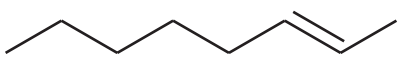
Context: part (d) (introduction)

(d) $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ is heated with **D** to produce three different molecules, **E**, **F** and **G**.**Fig. 4.2**

(e) (i) Both σ and π bonds are present in a molecule of **E** as a result of different types of hybridisation in the carbon atoms.

Complete Table 4.2 to show the number of carbon atoms with each type of hybridisation in a molecule of **E**.

Table 4.2

	number of carbon atoms		
	sp hybridised	sp ² hybridised	sp ³ hybridised
E 			

[2]

(ii) Describe the essential feature of an unbranched hydrocarbon that causes its molecules to show stereoisomerism. Explain how this feature leads to stereoisomerism.

[3]

46. M/J 24/P23/Q4(e)(iii)

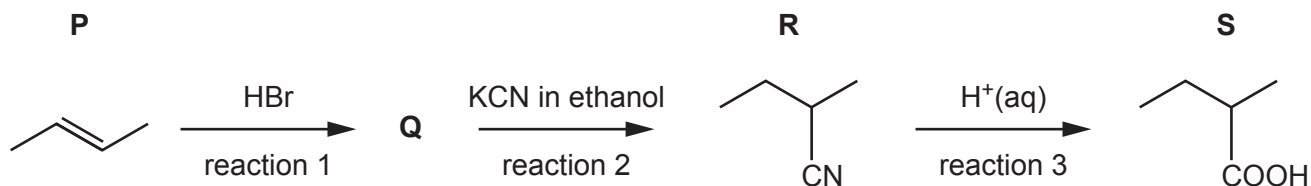
(iii) Give the systematic name for the organic product $\text{CH}_3\text{CH}(\text{NH}_2)\text{C}_2\text{H}_5$.

[1]

47. MAR 24/P22/Q4(a)

Context

- 4** Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

**Fig. 4.1**

- (a) **P**, **Q**, **R** and **S** show stereoisomerism.

Complete Table 4.1 by identifying with a tick (✓) the type of stereoisomerism that each molecule shows.

The type of stereoisomerism shown by **Q** is given.

Table 4.1

	P	Q	R	S
geometrical isomerism				
optical isomerism		✓		

[2]

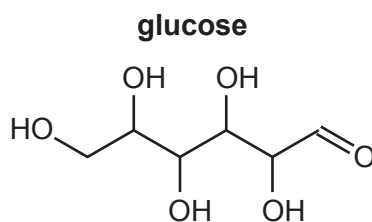
48. O/N 23/P21/Q3(e)(iii)

Context

- 3** Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

Context: part (e) (introduction)

- (e) The structure of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is shown in Fig. 3.2.

**Fig. 3.2**

- (iii) There are many structural isomers of $\text{C}_6\text{H}_{12}\text{O}_6$.

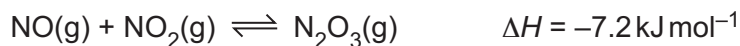
Define structural isomers.

[1]

49. O/N 23/P22/Q2(e)(i)

Context

- 2** NO and NO₂ react at 25 °C to give N₂O₃ as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

Context: part (e) (introduction)

- (e)** NO₂ is a brown gas that can be used to form nitric acid.

- (i)** NO₂ is a free radical.

Define free radical.

[1]

50. O/N 23/P22/Q4(a)(i)-(ii)

Context

- 4** Lactic acid, CH₃CH(OH)COOH, and pyruvic acid, CH₃COCOOH, both contain two functional groups.

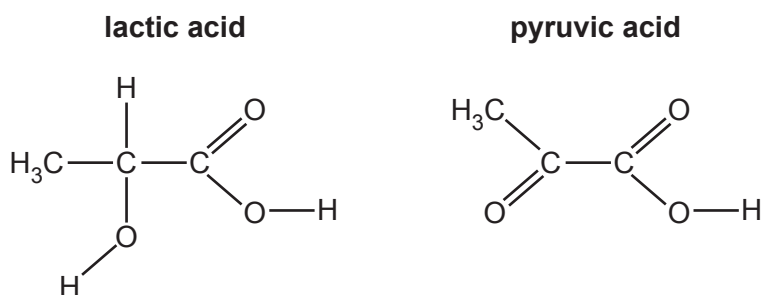


Fig. 4.1

- (a) (i)** Explain why lactic acid exists as optical isomers.

[1]

- (ii)** Give the systematic name of lactic acid.

[1]

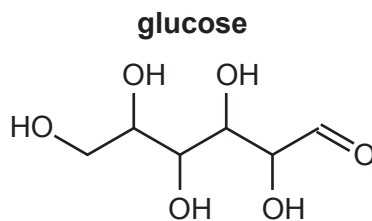
51. O/N 23/P23/Q3(e)(iii)

Context

3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

Context: part (e) (introduction)

(e) The structure of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is shown in Fig. 3.2.



(iii) There are many structural isomers of $\text{C}_6\text{H}_{12}\text{O}_6$.

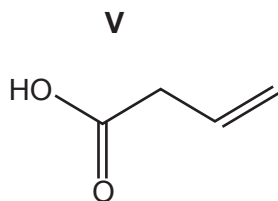
Define structural isomers.

[1]

52. M/J 23/P21/Q4(b)(iii)

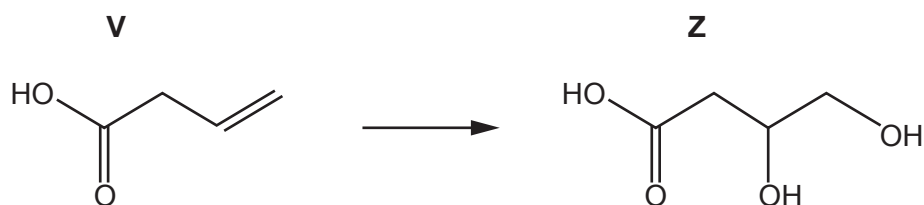
Context

4 **V** is a colourless liquid.



Context: part (b) (introduction)

(b) **V** reacts to form **Z** in a single reaction, as shown in Fig. 4.2.



(iii) Complete Table 4.1 to show the number of sp^2 and sp^3 hybridised carbon atoms that are present in a molecule of **V**.

Table 4.1

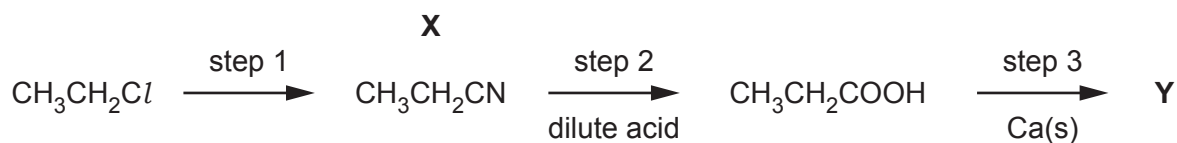
type of hybridisation	sp^2	sp^3
number of carbon atoms in V		

[2]

53. M/J 23/P22/Q3(a)(ii)

Context

- 3** Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from $\text{CH}_3\text{CH}_2\text{Cl}$.

**Fig. 3.1**

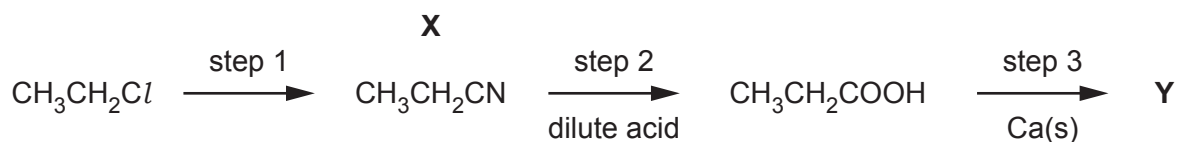
- (ii) Give the systematic name of **X**.

[1]

54. M/J 23/P22/Q3(c)

Context

- 3** Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from $\text{CH}_3\text{CH}_2\text{Cl}$.

**Fig. 3.1**

- (c) Complete Table 3.1 to show the number of sigma bonds (σ) and pi bonds (π) present in a molecule of **X**.

Table 3.1

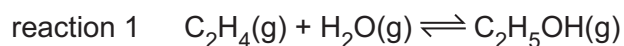
type of bond	number of bonds in X
sigma (σ)	
pi (π)	

[2]

55. M/J 23/P22/Q4(d)

Context

- 4** In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



- (d)** Describe the covalent bonds present between the carbon atoms in an ethene molecule by completing Table 4.2.

Table 4.2

	sigma (σ)	pi (π)
type of orbitals involved in bond		
how the orbitals overlap		

[2]

56. M/J 23/P22/Q5(a)

- 5** **(a)** Describe structural isomerism.

[1]

57. M/J 23/P22/Q5(c)

(c) **C** is a structural isomer of **A** and **B**.

C is straight chained and has two functional groups.

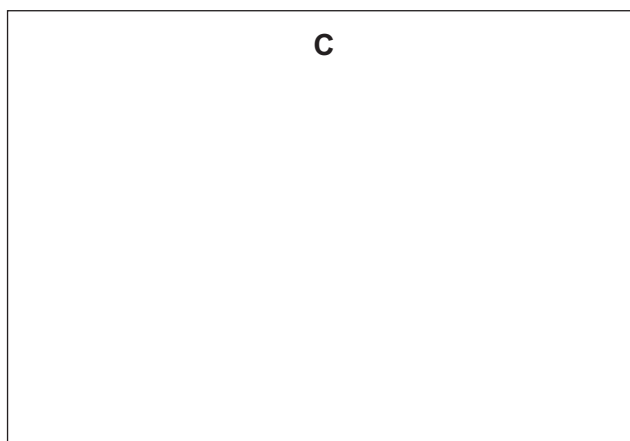
C shows only one type of stereoisomerism.

Table 5.2 describes observations when separate samples of **C** are added to different reagents.

Table 5.2

reagent	C
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)	no reaction
Br ₂ (aq)	orange to colourless
alkaline I ₂ (aq)	yellow precipitate appears

(i) Draw the structure of **C** in the box.



[2]

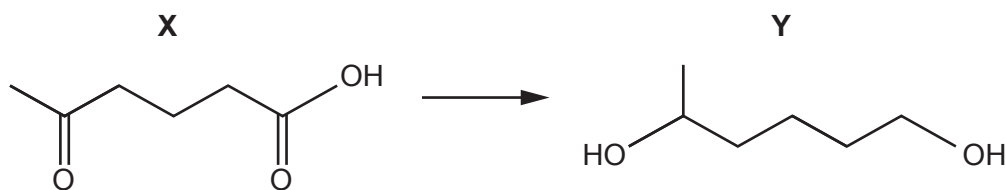
(ii) Name the type of stereoisomerism shown by molecules of **C**.

[1]

58. M/J 23/P23/Q5(c)

Context

5 **Y** is formed from **X** in a single-step reaction, as shown in Fig. 5.1.

**Fig. 5.1**

(c) Complete Table 5.1 to show the number of sp^2 and sp^3 hybridised carbon atoms in a molecule of **X**.

Table 5.1

type of hybridisation	sp^2	sp^3
number of carbon atoms in X		

[2]

59. M/J 23/P23/Q6(c)

Context

6 Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

(c) Molecules of **W**, $\text{CH}_2=\text{CHCN}$, do **not** show stereoisomerism.

(i) Describe stereoisomerism.

[1]

(ii) Describe the **two** essential features of an alkene molecule that cause it to show geometrical stereoisomerism.

[2]

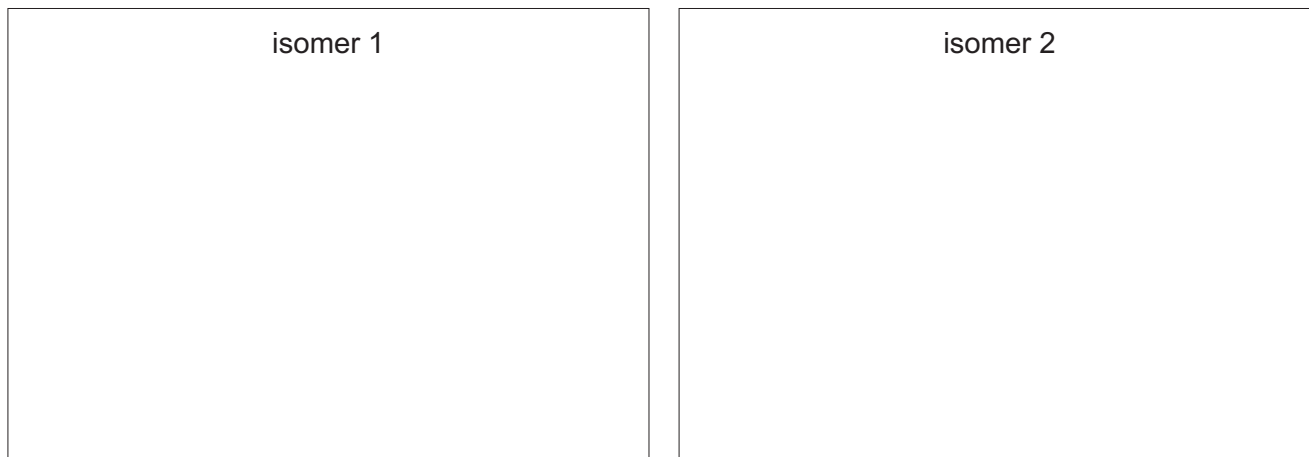
60. M/J 23/P23/Q6(d)

Context

6 Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

(d) Molecules of $\text{CH}_3\text{CH}(\text{OH})\text{CN}$ exist as a pair of optical isomers.

Draw three-dimensional diagrams in the boxes to show the optical isomers of $\text{CH}_3\text{CH}(\text{OH})\text{CN}$.



[1]

61. MAR 23/P22/Q3(a)(i)-(iii)

Context

3 Alkenes undergo an addition reaction with a 1:1 mixture of CO and H_2 to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H_2 .

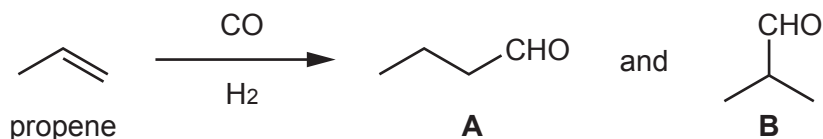


Fig. 3.1

(a) (i) Define addition reaction.

[1]

(ii) Aldehydes **A** and **B** are structural isomers.

State the type of structural isomerism shown by **A** and **B**.

[1]

(iii) Name **A**.

[1]

62. MAR 23/P22/Q3(c)

Context

- 3** Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

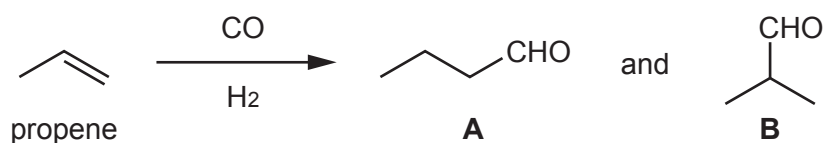


Fig. 3.1

- (c)** When propene reacts with CO and an excess of H₂, an alkane and a mixture of alcohols are formed instead. The alcohols are isomers of each other.

Suggest the molecular formulae of the alkane and the alcohols that are formed under these conditions.

molecular formula of alkane

molecular formula of alcohols

[2]

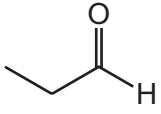
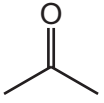
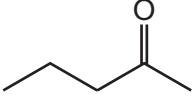
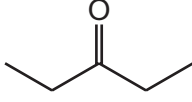
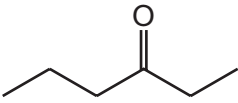
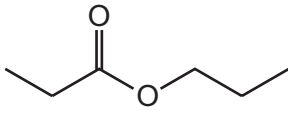
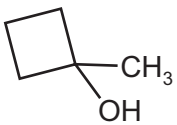
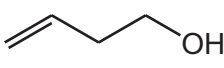
63. O/N 22/P22/Q3(a)(ii)-(iv)

Context

3 Organic compounds can be distinguished using chemical tests and analytical techniques.

Context: part (a) (introduction)

(a) Table 3.1 shows four pairs of organic compounds.**Table 3.1**

organic compounds		reagent	positive result of chemical test on identified compound
A1 	A2 		
B1 	B2 		
C1 	C2 		
D1 	D2 		

(ii) **A1** and **A2** are structural isomers.

Define structural isomers.

[1]

(iii) Give the systematic name of **B2**.

[1]

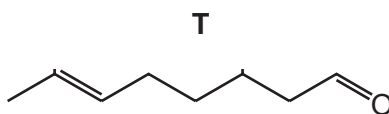
(iv) Deduce the molecular formula of **D1**.

[1]

64. M/J 22/P21/Q3(a)

Context

- 3** Liquids that contain molecules of **T** smell like lemons.

**Fig. 3.1**

- (a)** Molecules of **T** exist as a pair of stereoisomers.

Name the type of stereoisomerism shown by molecules of **T**. Explain your answer.

[2]

65. M/J 22/P21/Q4(c)(iii)

Context

4 Compound **V** is a liquid.**V** contains 77.2% carbon, 11.4% hydrogen and 11.4% oxygen by mass.**V** has a relative molecular mass of 280.

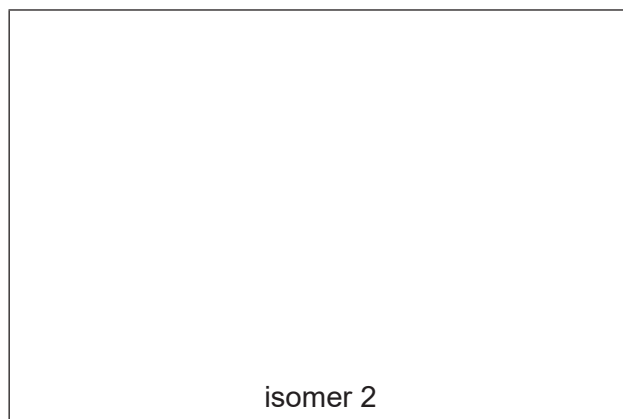
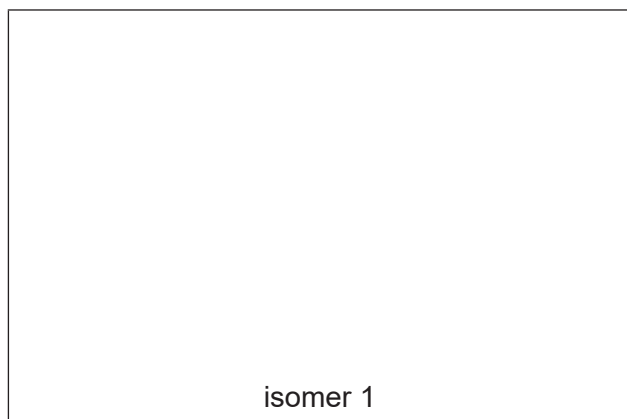
Context: part (c) (introduction)

(c) **W**, **X** and **Y** have the same molecular formula, $C_5H_{10}O$.**W**, **X** and **Y** are added separately to different reagents. Observations for these reactions are described in Table 4.1.**Table 4.1**

	+ 2,4-dinitrophenylhydrazine	+ alkaline $I_2(aq)$	+ Fehling's reagent and warm
W	orange precipitate seen	no change	orange-red precipitate seen
X	orange precipitate seen	yellow precipitate seen	no change
Y	orange precipitate seen		

(iii) **W** could be one of four structural isomers.

- Draw the skeletal formulae for two possible structural isomers of **W**.
- Describe the type of structural isomerism shown.



type of structural isomerism

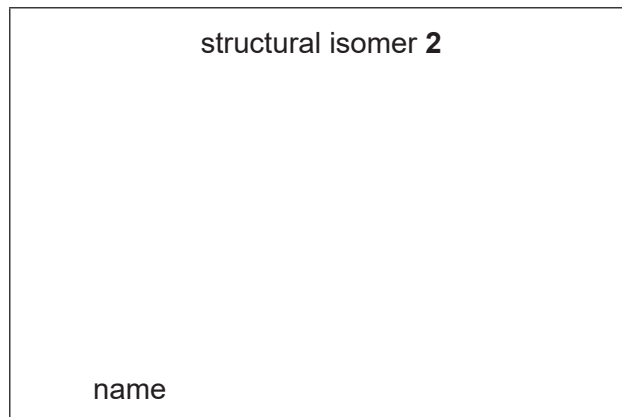
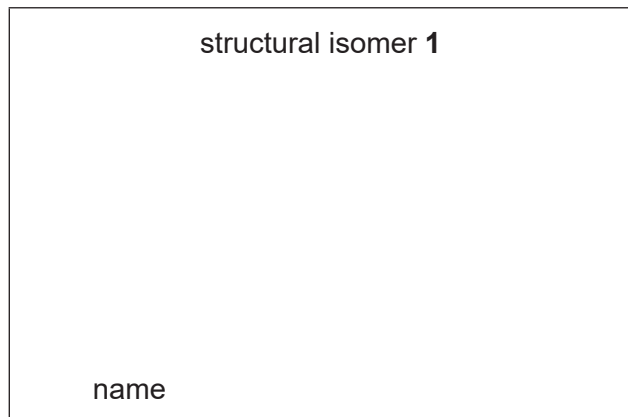
[3]

66. M/J 22/P22/Q4(a)

- 4 (a) Bromine reacts with butane in the presence of ultraviolet light to form bromobutane.

Two structural isomers with the molecular formula C_4H_9Br are produced during this reaction.

- (i) Draw the two structural isomers and state the systematic name of each isomer.



[2]

- (ii) Identify the type of structural isomerism shown in (a)(i).

[1]

67. M/J 22/P22/Q4(b)(i)-(ii)

Context: part (b) (introduction)

- (b) Halothane is an anaesthetic.

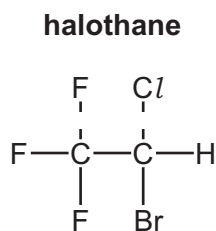


Fig. 4.1

- (i) Identify the chiral centre in halothane and mark it with an asterisk (*). [1]

When halothane reacts in ultraviolet light, homolytic fission occurs and the C–Br bond is broken.

- (ii) Construct an equation to show the homolytic fission of halothane, $CF_3CHBrCl$.

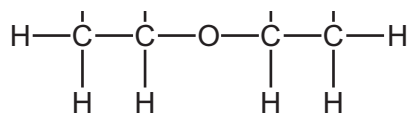
[1]

68. M/J 22/P23/Q3(b)

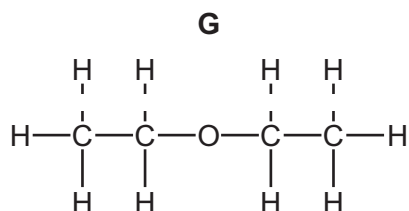
Context

- 3 G belongs to a group of compounds called ethers.



**Fig. 3.1**

- (b) The functional group in **G** is an oxygen atom bonded to two carbon atoms.

**Fig. 3.3**

G, **H** and **J** are structural isomers with molecular formula $\text{C}_4\text{H}_{10}\text{O}$.

H and **J** are straight chain molecules.

Table 3.3 shows the boiling points and reactions of **G**, **H** and **J** when heated under reflux with excess acidified $\text{K}_2\text{Cr}_2\text{O}_7$.

Table 3.3

	G	H	J
boiling point/ $^{\circ}\text{C}$	35	117	98
heat under reflux with excess acidified $\text{K}_2\text{Cr}_2\text{O}_7$	remains orange	orange to green	orange to green

- (i) Identify the type of structural isomerism shown between **G** and **H** using the information in Table 3.3.

[1]

- (ii) Identify the type of structural isomerism shown between **H** and **J** using the information in Table 3.3.

[1]

- (iii) Draw a possible structure for **H** and for **J**. State the systematic name for each structure.

H

name

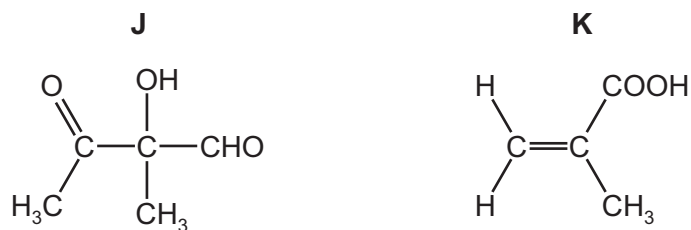
J

name

[3]

69. MAR 22/P22/Q4(a)(ii)

Context

4 Compounds **J** and **K** are found in plant oils.**Fig. 4.1****(ii)** **J** has **two** optical isomers.Draw the three-dimensional structures of the **two** optical isomers of **J**.

[2]

Answers Section

1. M/J 26/P21/Q5(a)

5(a)	compound	type of hybridisation of carbon atoms	2	M1 A sp ³ sp ₃ SP ³ etc. M2 A sp ² sp ₂ SP ² etc.
	1,2 dichloroethane	sp ³		
	chloroethene	sp ²		

2. M/J 26/P21/Q5(c)

5(c)	One atom takes both of the electrons when the covalent bond is broken ✓✓ M1 when a (covalent) bond / it is broken M2 one atom takes both of the electrons from the shared pair	2 A breaking = splitting = separation A breaking of a specific covalent bond e.g. when a CH bond is broken shared pair of electrons = covalent bond M2 A 'one atom takes both electrons' if reference to process involving breaking a (covalent) bond is described A takes = receives A electrons moves to / electrons transferred to I attract / pull towards M2 A species = atom R element, molecule, compound I radical I uneven number of electrons I formation of (positive and negative) ions I both electrons unequally distributed to one atom I formation of nucleophile and electrophile
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3. M/J 26/P22/Q5(a)

5(a)	structure of product	systematic name of product	2	Max 1 for consistently incorrect alkane stem that has correct idea of 1-chloro and 2-chloro. Max 1 6-chlorohexane and 5-chlorohexane. Max 1 for consistent errors e.g. 1-chlorohexane and 2-chlorohexane OR 1-chlorohexane and 2-clorohexane OR chloro-1-hexane and chloro-2-hexane A missing hyphen
	CH ₃ (CH ₂) ₅ Cl	1-chlorohexane		
	CH ₃ (CH ₂) ₃ CHClCH ₃	2-chlorohexane		
	CH ₃ (CH ₂) ₃ CH ₂ ClCH ₂ Cl	1,2 dichlorohexane		
	CH ₃ (CH ₂) ₁₀ CH ₃	dodecane		
	CH ₃ (CH ₂) ₃ CH(CH ₃)CH(CH ₃)(CH ₂) ₃ CH ₃	5,6-dimethyldecane		

4. M/J 26/P22/Q5(b)(i)-(ii)

5(b)(i)	(species) with one or more unpaired electrons	1 BOD species with (an) unpaired electron(s) I lone / single / free / odd number of electrons (in valence shell) I species with specified number (except 1) unpaired electrons BOD species with one or two unpaired electrons R it is an unpaired electron BOD species = atom, ion, particle, molecule, substance, element, compound
5(b)(ii)	homolytic (fission)	1 BOD homolytical A homolysis I covalent I photolytic

5. M/J 26/P22/Q6(a)

6(a)(i)		1	A any unambiguous cis / Z structure of P I connectivity I trans structure
6(a)(ii)	M1 limited / restricted / lack of / no rotation about the pi bond / π bond of C=C M2 (two) different groups attached to each C of the C=C / each end of the C=C	2	M1 A limited / restricted / lack of / no rotation about the C=C reference to C=C needs to be made once only BOD C–C double bond I double bond (there is a C=O also in the molecule) I movement rotate = twist = spin R (compound) P has a chiral carbon M2 each = both A different atoms / functional groups / H and COOH / species I each side (of the C=C) R different compounds / molecules

6. M/J 26/P22/Q6(b)(ii)

6(b)(ii)	M1 M2 	2	COOH = CO ₂ H A C ₂ H ₃ O ₂ M1 4 correct groups around central C in any orientation (might be seen on 'isomer 2') M2 enantiomer of their M1 structure (does not have to be shown as its mirror image). M2 R bonds in the plane that are opposite to each other I bond shapes that are not associated with the chiral carbon. I bond angles I connectivity
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7. M/J 26/P23/Q4(c)(i)-(ii)

4(c)(i)	M1 (molecules / isomers with) the same molecular formula / same number of atoms of each element M2 different arrangement of atoms	2	A similar = same M1 A same molecular formula I same chemical formulas I same / similar molecular number M2 A different structures M2 A different structural formulas / different displayed formulas / different skeletal formulas I arrangement of bonds / different arrangements in space I different carbon skeleton / functional groups / chain
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4(c)(ii)	M1 compound W AND compound V M2 both have (2) chiral centres / (2) asymmetric carbon atoms	2 M1 compound W AND compound V ✓ M2 one or two chiral centre on V & W ✓ M1 compound V OR compound W M2 show one or two chiral centre on the isomer identified ^✓ R molecules / compounds for atoms / groups R C attached to 4 different alkyl groups M2 A they have a chiral centre / an asymmetric carbon (atom) chiral = 4 different groups / atoms bonded to a carbon M2 may be indicated by annotation on Figure 4.2 e.g. $\text{CH}_2\text{OHCH}_2^*\text{CH}(\text{OH})$ and / or $^*\text{CH}(\text{OH})\text{CH}_2\text{OH}$
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8. M/J 26/P23/Q5(a)(i)

5(a)(i)	structure of product	systematic name of product	2	A any unambiguous structures
	$\text{CH}_3(\text{CH}_2)_2\text{Cl}$	1 chloropropane		
	$\text{CH}_3\text{CHCl} / \text{CH}_3$	2-chloropropane		
	$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2,4-dimethylbutane		
	$\text{CH}_3(\text{CH}_2)_4\text{CH}_3$	hexane		
	$\text{CH}_3(\text{CH}_2)_2\text{CH}(\text{CH}_3)_2$	2-methylpentane		

9. M/J 26/P24/Q5(c)

5(c)	positional	1 A regoisomerism A position (isomer)
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10. MAR 26/P22/Q2(a)(i)

2(a)(i)	(free) radical	1
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11. MAR 26/P22/Q3(b)

3(b)(i)		1
3(b)(ii)	M1 M2 <i>trans</i>-but-2-ene	2
3(b)(iii)	M1 (two) different groups attached to each carbon of $\text{C}=\text{C}$ M2 restricted / lack of / no rotation about double / $\text{C}=\text{C}$ / π / π bond	2

12. MAR 26/P22/Q4(d)

4(d)(i)	condensation	1
4(d)(ii)	methanal	1
4(d)(iii)	$\text{C}_6\text{H}_7\text{NO}_2$	1
4(d)(iv)		1

4(d)(v)	• $\times$ on absorption at $\sim 1550\text{ cm}^{-1}$ • $\text{C}=\text{C}$ • (peak / $\text{C}=\text{C}$ / bond) disappears (during / after polymerisation)	2
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13. O/N 25/P21/Q3(b)

3(b)	two of: 	2
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14. O/N 25/P21/Q3(c)(i)

3(c)(i)	homolytic	1
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15. O/N 25/P21/Q4(b)(ii)

4(b)(ii)	M1 (molecules that have the) same structural formula (and same molecular formula) M2 but different 3D / spatial arrangement of atoms / groups	2
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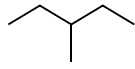
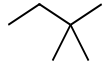
16. O/N 25/P21/Q4(c)(i)

4(c)(i)	2,3-dimethylbut-1-ene	1
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17. O/N 25/P22/Q3(a)

3(a)	M1 (family of molecules / compounds / series with) same general formula OR (family of molecules where) molecular formula differs (only) by (units of) $\text{CH}_2(n)$. M2 similar chemical properties / reactions	2
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18. O/N 25/P22/Q3(d)(i)

3(d)(i)	M1  M2 	2
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19. O/N 25/P22/Q4(a)(iii)-(v)

4(a)(iii)	homolytic	1
4(a)(iv)	1,2-dichloropropane	1
4(a)(v)	M1 (type of structural isomerism) positional M2 (type of stereoisomerism) optical	2

20. O/N 25/P23/Q3(b)

3(b)	two of: 	2
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21. O/N 25/P23/Q3(c)(i)

3(c)(i)	homolytic	1
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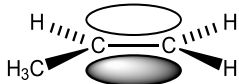
22. O/N 25/P23/Q4(b)(ii)

4(b)(ii)	M1 (molecules that have the) same structural formula (and same molecular formula) M2 but different 3D / spatial arrangement of atoms / groups	2
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23. O/N 25/P23/Q4(c)(i)

4(c)(i)	2,3-dimethylbut-1-ene	1
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24. O/N 25/P24/Q3(a)(iv)

3(a)(iv)		1
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25. M/J 25/P22/Q3(a)(i)

3(a)(i)	two (or more) reactants / molecules react to make (only) one product	1
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26. M/J 25/P22/Q4(c)(i)

4(c)(i)	sp ³	1
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27. M/J 25/P23/Q5(b)(i)

5(b)(i)	number of σ bonds = 14 number of π bonds = 2	2
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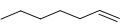
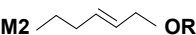
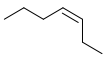
28. M/J 25/P23/Q6(a)(i)

6(a)(i)	two or more reactants / molecules react to make (only) one product	1
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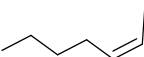
29. M/J 25/P24/Q4(a)

4(a)	C ₅ H ₁₀ O ₂	1
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30. M/J 25/P24/Q5(a)-(b)

5(a)	M1  M2  OR 	2
5(b)(i)	no rotation / restricted rotation of C=C / pi bond	1
5(b)(ii)	(W / it has two) different groups bonded to each C of the double bond	1

31. M/J 25/P24/Q5(c)

5(c)		1
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32. M/J 25/P24/Q6(a)(iii)

6(a)(iii)	alkene(s)	1
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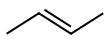
33. MAR 25/P22/Q1(c)(i)

1(c)(i)	contains a(/ one or more) C=C / double / multiple bonds	1
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34. MAR 25/P22/Q4(b)(iii)

4(b)(iii)	atom / carbon with four different atoms / (functional) groups attached / bonded	1
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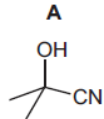
35. O/N 24/P21/Q1(d)(ii)-(iv)

1(d)(ii)	molecules with the same structural formula and same molecular formula with different 3D / different spatial arrangement of atoms / groups	1
1(d)(iii)	a single product is made	1
1(d)(iv)		1
		1

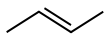
36. O/N 24/P21/Q2(b)(i)

2(b)(i)	a species with one or more unpaired electrons	1
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37. O/N 24/P21/Q4(a)(ii)

4(a)(ii)	A's central carbon atom is not bonded to four different atoms/groups of atoms 	1
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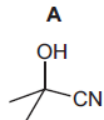
38. O/N 24/P23/Q1(d)(ii)-(iv)

1(d)(ii)	molecules with the same structural formula and same molecular formula with different 3D / different spatial arrangement of atoms / groups	1
1(d)(iii)	a single product is made	1
1(d)(iv)		1
		1

39. O/N 24/P23/Q2(b)(i)

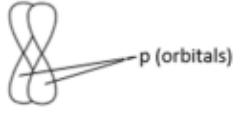
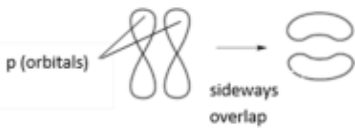
2(b)(i)	a species with one or more unpaired electrons	1
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40. O/N 24/P23/Q4(a)(ii)

4(a)(ii)	A's central carbon atom is not bonded to four different atoms/groups of atoms 	1
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41. M/J 24/P21/Q5(b)

5(b)(i)	sp^2	1
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5(b)(ii)	M1 σ bond – diagram showing how σ bond is made from direct overlap of orbitals i.e.  OR  M2 π bond - diagram showing how π bond is made from sideways overlap of 2$\times$ p-orbitals i.e.  OR  OR 	2
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42. M/J 24/P21/Q5(c)(i)

5(c)(i)	EITHER (pair of) electrons in π bond are further away from the nuclei so weaker attraction OR (pair of) electrons in σ bond are closer to the two nuclei so stronger attraction	1
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43. M/J 24/P21/Q6(a)

6(a)(i)	(molecules with the) same structural formula (and same molecular formula) with different arrangement of atoms / groups in space	1
6(a)(ii)	M1 number of stereoisomers = 8 M2 reasoning: 2 chiral centres AND 1 C=C (producing a cis-and trans or geometrical pair)	2
6(a)(iii)	C ₁₃ H ₂₀ O ₃	1
6(a)(iv)	ester AND carbonyl / ketone AND C=C (bond) / alkene	1

44. M/J 24/P22/Q4(a)

4(a)	Three-dimensional stereoisomers of CH₃(CH₂)₅CHBrCH₃ M1 correct representation of left-hand answer as 3d structure of one of the enantiomers of CH₃(CH₂)₅CHBrCH₃ M2 correct representation of 3d structure of the second enantiomer of CH₃(CH₂)₅CHBrCH₃	2
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45. M/J 24/P22/Q4(e)

4(e)(i)	<table><tr><td></td><td>0</td><td>2</td><td>6</td></tr></table> 3 correct answers = 2 marks 2 correct answers = 1 mark 0 or 1 correct answers = 0 marks		0	2	6	2
	0	2	6			
4(e)(ii)	M1 presence of a π bond M2 limited rotation of C=C bond M3 EITHER two different groups on each / both double bonded carbon atoms OR two different atoms on each / both double bonded carbon atom OR one H / atom and one alkyl group on each / both double bonded carbon atom	3				

46. M/J 24/P23/Q4(e)(iii)

4(e)(iii)	butan-2-amine OR butyl-2-amine	1
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47. MAR 24/P22/Q4(a)

4(a)	<table><tr><td>molecule</td><td>P</td><td>Q</td><td>R</td><td>S</td></tr><tr><td>geometrical</td><td>✓</td><td>(✗)</td><td>(✗)</td><td>(✗)</td></tr><tr><td>optical</td><td>(✗)</td><td>✓</td><td>✓</td><td>✓</td></tr></table>					molecule	P	Q	R	S	geometrical	✓	(✗)	(✗)	(✗)	optical	(✗)	✓	✓	✓	1
	molecule	P	Q	R	S																
	geometrical	✓	(✗)	(✗)	(✗)																
	optical	(✗)	✓	✓	✓																
	<table><tr><td>molecule</td><td>P</td><td>Q</td><td>R</td><td>S</td></tr><tr><td>geometrical</td><td>✓</td><td>(✗)</td><td>(✗)</td><td>(✗)</td></tr><tr><td>optical</td><td>(✗)</td><td>✓</td><td>✓</td><td>✓</td></tr></table>					molecule	P	Q	R	S	geometrical	✓	(✗)	(✗)	(✗)	optical	(✗)	✓	✓	✓	1
	molecule	P	Q	R	S																
geometrical	✓	(✗)	(✗)	(✗)																	
optical	(✗)	✓	✓	✓																	
CH ₃ CHBrCH ₂ CH ₃																					

48. O/N 23/P21/Q3(e)(iii)

3(e)(iii)	molecules with the same molecular formula but different structural formulae	1
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49. O/N 23/P22/Q2(e)(i)

2(e)(i)	species with 1 or more unpaired electrons	1
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50. O/N 23/P22/Q4(a)(i)-(ii)

4(a)(i)	(it has molecules that) contain a chiral carbon / centre OR are non-superimposable mirror images	1
4(a)(ii)	2-hydroxypropanoic acid	1

51. O/N 23/P23/Q3(e)(iii)

3(e)(iii)	molecules with the same molecular formula but different structural formulae	1
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52. M/J 23/P21/Q4(b)(iii)

4(b)(iii)	type of hybridisation	sp ²	sp ³	2
	number of carbon atoms in V	3	1	

53. M/J 23/P22/Q3(a)(ii)

3(a)(ii)	propanenitrile	1
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54. M/J 23/P22/Q3(c)

3(c)	type of bond in X	number of bonds in X	2
	sigma σ	8	
	pi π	2	

55. M/J 23/P22/Q4(d)

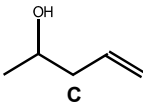
4(d)		sigma	pi	2
	type of orbitals involved in bond	$sp^2(-sp^2)$	$(2)p(-(2)p)$	
	how the orbitals overlap	direct	sideways	

Two correct for one mark, four correct for two marks.

56. M/J 23/P22/Q5(a)

5(a)	molecules / isomers with the same molecular formula / same number of atoms of each element AND different structural formulae / different structures	1
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57. M/J 23/P22/Q5(c)

5(c)(i)	 M1 $CH_3CH(OH)$ present M2 five carbons in a straight chain AND molecular formula $C_5H_{10}O$ AND 1 $C=C$ structure which shows only one type of stereoisomerism	2
5(c)(ii)	optical	1

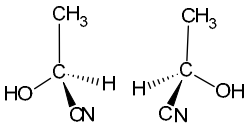
58. M/J 23/P23/Q5(c)

5(c)	type of hybridisation	sp^2	sp^3	2
	number of carbon atoms in X	2	4	

59. M/J 23/P23/Q6(c)

6(c)(i)	(molecules / compounds with the) same structural formula (and same molecular formula) with different arrangement of atoms / groups in space	1
6(c)(ii)	M1 restricted rotation of $C=C$ (bond) M2 each C of the double bond has two different groups / two different atoms (attached)	2

60. M/J 23/P23/Q6(d)

6(d)	 A pair of molecules of 2-hydroxypropanenitrile with bonds represented in 3-d AND the pair of molecules are non-superimposable images	1
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61. MAR 23/P22/Q3(a)(i)-(iii)

3(a)(i)	(a reaction where) two (or more) compounds / reagents / molecules form (only) one product	1
3(a)(ii)	position(al isomerism)	1
3(a)(iii)	butanal	1

62. MAR 23/P22/Q3(c)

3(c)	alkane	C_3H_8	1
	alcohols	$C_4H_{10}O$	1

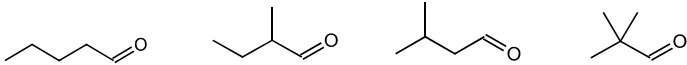
63. O/N 22/P22/Q3(a)(ii)-(iv)

3(a)(ii)	molecules with the same molecular formula AND different structural formulae <i>owtte</i>	1
3(a)(iii)	pentan-3-one	1
3(a)(iv)	<u>molecular</u> formula $C_5H_{10}O$	1

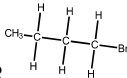
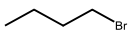
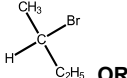
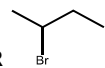
64. M/J 22/P21/Q3(a)

3(a)	M1 optical M2 one of the C atoms has 4 different groups / atoms attached	2
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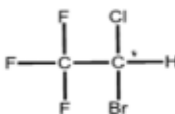
65. M/J 22/P21/Q4(c)(iii)

4(c)(iii)	M1 / M2  M3 chain (isomerism)	3
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66. M/J 22/P22/Q4(a)

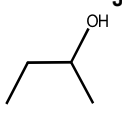
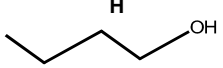
4(a)(i)	$CH_3(CH_2)_2CH_2Br$  OR  name: 1-bromobutane $CH_3CH_2CH(Br)CH_3$  OR  name: 2-bromobutane M1 structure of isomer 1 AND name M2 structure of isomer 2 AND name (in any order)	2
4(a)(ii)	positional	1

67. M/J 22/P22/Q4(b)(i)-(ii)

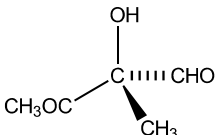
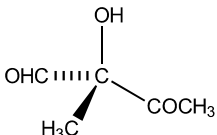
4(b)(i)		1
4(b)(ii)	$CF_3CBrClH \rightarrow CF_3CHCl(\bullet) + Br(\bullet)$	1

68. M/J 22/P23/Q3(b)

3(b)(i)	functional group	1
3(b)(ii)	positional	1

3(b)(iii)	 J name: butan-2-ol  H name: butan-1-ol M1 correct structures M2 correct structures in correct boxes H and J M3 correct corresponding name of both H and J	3
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69. MAR 22/P22/Q4(a)(ii)

4(a)(ii)	 	2
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